Proposed research title

Real-Time Multimodal EMG–IMU Control of a 6-DOF Robotic Arm: Performance, Responsiveness, and Reliability Evaluation

Primary research question

> To what extent does integrating eight-channel EMG signals with three-channel IMU data improve the performance, responsiveness, and reliability of real-time control of a six-degree-of-freedom robotic arm compared with EMG-only and IMU-only control?

Research objective

To develop and experimentally evaluate a real-time robotic-arm control system that combines EMG-based gesture recognition and IMU-based hand-orientation tracking, comparing its performance with single-modality control using recognition metrics, end-to-end latency, unintended movement frequency, and task completion performance.

Expected contribution

A reproducible evaluation of how combining muscle-activity and hand-orientation signals affects robotic-arm control, including the trade-offs between responsiveness, reliability, and task performance.